



TOSHKENT SHAHRIDAGI INHA UNIVERSITETI INHA UNIVERSITY IN TASHKENT

STUDENT NAME:

STUDENT ID NUMBER:

FINAL EXAMINATION — Fall 2022

COURSE NAME:

Circuit & Lab

COURSE NUMBER:

CSE2120

EXAMINATION DATE:

00-12-2022

TIME:

EXAMINATION DURATION:

60 minutes

ADDITIONAL MATERIALS
ALLOWED TO USE:

Calculators are NOT allowed.

(Do not use)	(Do not use)	(Do not use)	(Do not use)	(Do not use)

Please do not open the examination paper until directed to do so.

READ INSTRUCTIONS FIRST:

Desks should be free from all unnecessary items (books, notes, technology, food, water, clothes)

Use of any electronic device (Phone, iPod, iPad, laptop) is not allowed during the examination.

Cheating, talking to fellow students, singing, turning back are not allowed

Write ID number in each page of your examination paper.

All answers should be written in the space provided for each question, unless specified the other way.

If you have a problem please raise your hand and wait quietly for a Proctor.

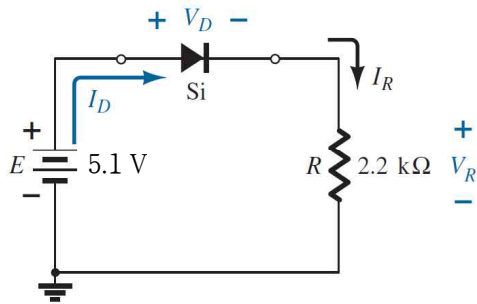
You are not allowed to leave the exam room until you submit the exam papers.

ID () Name ()

Answer Table

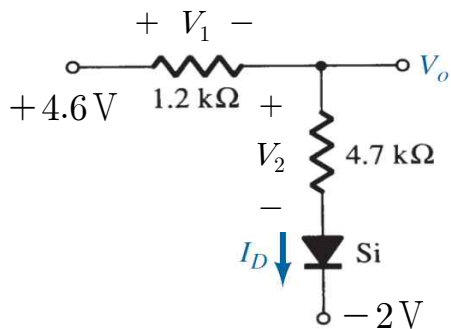
1	2	3	4	5	6	7	8	9	10	11	12	13
14	15	16	17	18	19	20	21	22	23	24	25	26

[1-3] For the given Si diode circuit,



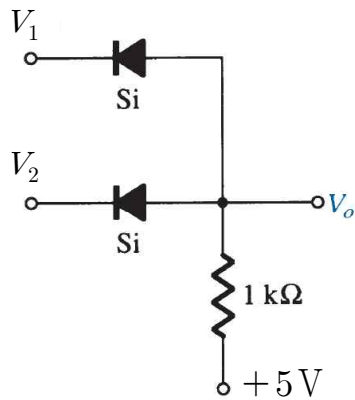
- Determine V_D .
 (1) 0.7 V (2) 0.2 V (3) 5.1 V (4) 0 V (5) 1.2 V
- Determine V_R .
 (1) -5.1 V (2) 5.1 V (3) 0.7 V (4) 4.4 V (5) 0.2 V
- Determine I_D .
 (1) 0 mA (2) 2 mA (3) 2.3 mA (4) 3 mA (5) 5.1 mA

[4-7] For the given Si diode circuit,



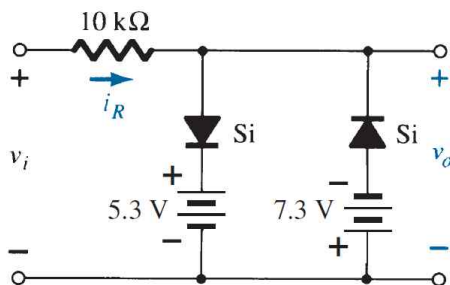
- Determine V_1 .
 (1) 0.7 V (2) 1.2 V (3) 2 V (4) 4.6 V (5) 4.7 V
- Determine V_2 .
 (1) -2 V (2) 0.2 V (3) 2.6 V (4) 4.6 V (5) 4.7 V
- Determine V_o .
 (1) -1.3 V (2) 0.7 V (3) 1.2 V (4) 3.4 V (5) 3.9 V
- Determine I_D .
 (1) 0 A (2) 0.43 mA (3) 1 mA (4) 2.43 mA (5) 3.8 mA

[8-10] Determine V_o .



8. When $V_1 = 0$ V and $V_2 = 0$ V , find V_o .
 (1) 4.3 V (2) 0.7 V (3) 1.4 V (4) 5 V (5) 0 V
9. When $V_1 = 0$ V and $V_2 = 7$ V , find V_o .
 (1) 0.7 V (2) 5 V (3) 4.3 V (4) 0 V (5) 7 V
10. When $V_1 = 7$ V and $V_2 = 7$ V , find V_o .
 (1) 0.7 V (2) 6.3 V (3) 5 V (4) 7 V (5) 4.3 V

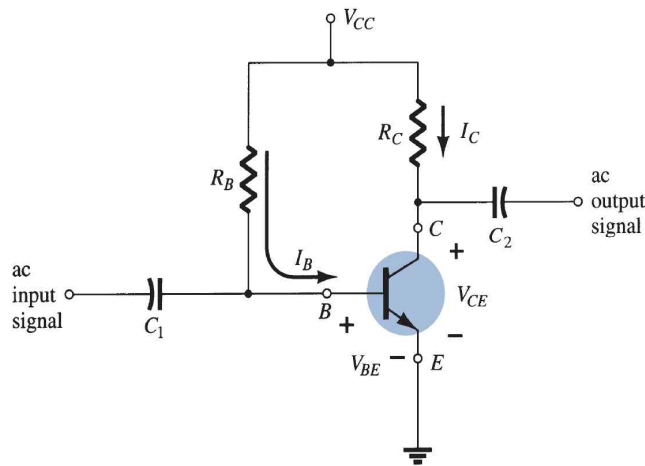
[11-15] For the given diode circuit,



11. When $V_i = 10$ V, find V_o .
 (1) -8 V (2) 10 V (3) -4.6 V (4) 5.3 V (5) 6 V
12. When $V_i = 5$ V, find V_o .
 (1) -6 V (2) -4.6 V (3) 0.7 V (4) 5 V (5) 6 V
13. When $V_i = 0$ V, find V_o .
 (1) 0 V (2) 2 V (3) 0.7 V (4) 5.3 V (5) 7.3 V
14. When $V_i = -5$ V, find V_o .
 (1) -5 V (2) -4.6 V (3) 0 V (4) 5.3 V (5) -6.6 V
15. When $V_i = -10$ V, find V_o .
 (1) 4.7 V (2) -8 V (3) -6.6 V (4) -10 V (5) 2.7 V

[16-19] For the fixed-bias configuration of a Si Bipolar Junction Transistor ;

$$(V_{cc} = 10.7 \text{ V} , R_B = 200 \text{ k}\Omega , R_C = 1 \text{ k}\Omega , \beta = 100)$$



16. Determine I_B .

- (1) $10 \mu\text{A}$ (2) $50 \mu\text{A}$ (3) $100 \mu\text{A}$ (4) $500 \mu\text{A}$ (5) 1 mA

17. Determine V_{BE} .

- (1) 0 V (2) 0.2 V (3) 0.7 V (4) 1.2 V (5) 10 V

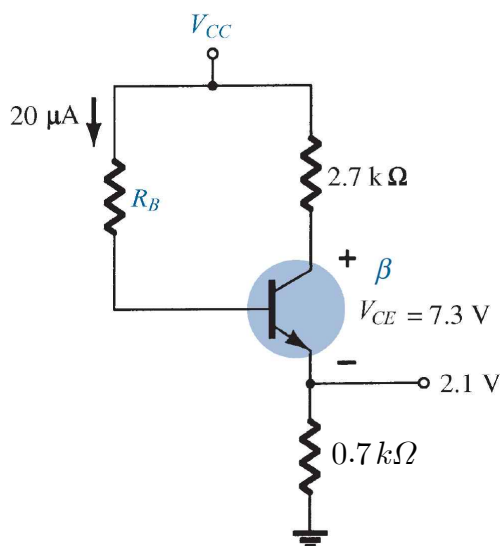
18. Determine I_C .

- (1) 0 A (2) 0.98 mA (3) 1 mA (4) 2.5 mA (5) 5 mA

19. Determine V_{CE} .

- (1) 0 V (2) 0.7 V (3) 5 V (4) 5.7 V (5) 10 V

[20-22] Given the information provided in the Si BJT circuit:



20. Determine β . (Choose the nearest, please.)

- (1) 50 (2) 100 (3) 150 (4) 200 (5) 250

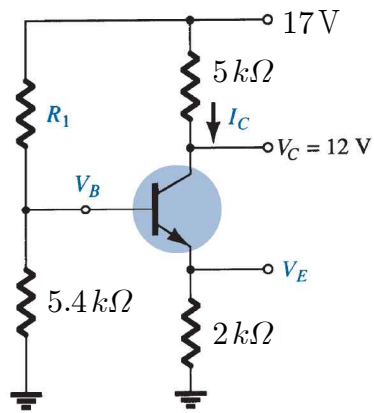
21. Determine V_{CC} . (Choose the nearest, please.)

- (1) 7.3 V (2) 9.4 V (3) 14.1 V (4) 17.5 V (5) 20.9 V

22. Determine R_B . (Choose the nearest, please.)

- (1) $670 \text{ k}\Omega$ (2) $70 \text{ k}\Omega$ (3) $105 \text{ k}\Omega$ (4) $125 \text{ k}\Omega$ (5) $735 \text{ k}\Omega$

[23-26] Given the information provided in the Si BJT circuit: [Approximation Method]



23. Determine I_C .
 (1) 0 A (2) 0.98 mA (3) 1 mA (4) 1.12 mA (5) 3.8 mA
24. Determine V_E .
 (1) 2 V (2) 2.7 V (3) 0.7 V (4) 12 V (5) 17 V
25. Determine V_B .
 (1) 5.3 V (2) 12.7 V (3) 0.7 V (4) 11.3 V (5) 2.7 V
26. Determine R_1 .
 (1) 1.55 $k\Omega$ (2) 2.78 $k\Omega$ (3) 10.4 $k\Omega$ (4) 28.6 $k\Omega$ (5) 45.1 $k\Omega$